pns_c45

November 14, 2025

[37]:
```
import c45
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.preprocessing import LabelEncoder, OneHotEncoder
```

[38]:
```
modelo = c45.C45Classifier()
```

[48]:
```
df = pd.read_parquet('dados_filtradoseprocessados_v1.parquet')
```

[49]:
```
df = df.drop(['N008', 'P02801', 'M007', 'P051', 'P036', 'P038', 'P04001',␣
 ↪'P039', 'M005010'], axis=1)
df.info()
encoder = OneHotEncoder()

categorical_cols = df.select_dtypes(include=['object', 'category']).columns

df_encoded = pd.get_dummies(df, drop_first=True)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 53139 entries, 0 to 53138
Data columns (total 23 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   A001     53139 non-null  object
 1   C001     53139 non-null  float64
 2   C006     53139 non-null  object
 3   C008     53139 non-null  float64
 4   C009     53139 non-null  object
 5   C011     53139 non-null  object
 6   D001     53139 non-null  object
 7   M01601   53139 non-null  object
 8   N010     53139 non-null  object
 9   N011     53139 non-null  object
 10  N016     53139 non-null  object
 11  N017     53139 non-null  float64
 12  P02002   53139 non-null  float64
```

```
 13  P02601   53139 non-null  float64
 14  P02602   53139 non-null  float64
 15  P027     53139 non-null  object
 16  P034     53139 non-null  object
 17  P044     53139 non-null  object
 18  P046     53139 non-null  object
 19  P050     53139 non-null  object
 20  Q00201   53139 non-null  object
 21  VDD004A  53139 non-null  object
 22  VDF004   53125 non-null  object
dtypes: float64(6), object(17)
memory usage: 9.3+ MB
```

[52]:
```
df_encoded.head()
df_encoded.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 53139 entries, 0 to 53138
Data columns (total 52 columns):
 #   Column                                        Non-Null Count  Dtype
---  ------                                        --------------  -----
 0   C001                                          53139 non-null  float64
 1   C008                                          53139 non-null  float64
 2   N017                                          53139 non-null  float64
 3   P02002                                        53139 non-null  float64
 4   P02601                                        53139 non-null  float64
 5   P02602                                        53139 non-null  float64
 6   A001_Casa                                     53139 non-null  bool
 7   A001_Habitacao em casa de comodos (cortico)   53139 non-null  bool
 8   C006_Mulher                                   53139 non-null  bool
 9   C009_Branca                                   53139 non-null  bool
 10  C009_Ignorado                                 53139 non-null  bool
 11  C009_Indigena                                 53139 non-null  bool
 12  C009_Parda                                    53139 non-null  bool
 13  C009_Preta                                    53139 non-null  bool
 14  C011_Divorciado/separado                      53139 non-null  bool
 15  C011_Solteiro                                 53139 non-null  bool
 16  C011_Viuvo                                    53139 non-null  bool
 17  D001_Sim                                      53139 non-null  bool
 18  M01601_De 2 a 3 vezes por mes                 53139 non-null  bool
 19  M01601_Mais de uma vez                        53139 non-null  bool
 20  M01601_Nenhuma vez                            53139 non-null  bool
 21  M01601_Uma vez no ano                         53139 non-null  bool
 22  M01601_Uma vez por mes                        53139 non-null  bool
 23  N010_Menos da metade dos dias                 53139 non-null  bool
 24  N010_Nenhum dia                               53139 non-null  bool
 25  N010_Quase todos os dias                      53139 non-null  bool
 26  N011_Menos da metade dos dias                 53139 non-null  bool
```

```
 27  N011_Nenhum dia                                53139 non-null  bool
 28  N011_Quase todos os dias                       53139 non-null  bool
 29  N016_Menos da metade dos dias                  53139 non-null  bool
 30  N016_Nenhum dia                                53139 non-null  bool
 31  N016_Quase todos os dias                       53139 non-null  bool
 32  P027_Nao bebo nunca                            53139 non-null  bool
 33  P027_Uma vez ou mais por mes                   53139 non-null  bool
 34  P034_Sim                                       53139 non-null  bool
 35  P044_Sim                                       53139 non-null  bool
 36  P046_Sim                                       53139 non-null  bool
 37  P050_Sim / Já fumou                            53139 non-null  bool
 38  P050_Sim/ Já fumou                             53139 non-null  bool
 39  Q00201_Sim                                     53139 non-null  bool
 40  VDD004A_Fundamental incompleto                 53139 non-null  bool
 41  VDD004A_Medio completo                         53139 non-null  bool
 42  VDD004A_Medio incompleto                       53139 non-null  bool
 43  VDD004A_Sem instrucao                          53139 non-null  bool
 44  VDD004A_Superior completo                      53139 non-null  bool
 45  VDD004A_Superior incompleto                    53139 non-null  bool
 46  VDF004_Entre 1 e 2 salarios minimos            53139 non-null  bool
 47  VDF004_Entre 1/2 e 1 salario minimo            53139 non-null  bool
 48  VDF004_Entre 1/4 e 1/2 salario minimo          53139 non-null  bool
 49  VDF004_Entre 2 e 3 salarios minimos            53139 non-null  bool
 50  VDF004_Entre 3 e 5 salarios minimos            53139 non-null  bool
 51  VDF004_Mais de 5 salarios minimos              53139 non-null  bool
dtypes: bool(46), float64(6)
memory usage: 4.8 MB
```

[ ]:
```
x = df_encoded.drop('Q00201_Sim', axis=1)
y = df_encoded['Q00201_Sim']
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)


modelo.fit(x_train, y_train)
y_pred = modelo.predict(x_test)
```

[54]:
```
y_pred
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
[[4227 2293]
 [2177 1931]]
              precision    recall  f1-score   support

       False       0.66      0.65      0.65      6520
        True       0.46      0.47      0.46      4108
```

```
              accuracy                           0.58     10628
             macro avg       0.56      0.56      0.56     10628
          weighted avg       0.58      0.58      0.58     10628
```

[ ]: